Global Semiconductors and Semiconductor Capital Equipment

Global Semis: June 2026 WSTS Tracker - Sales +10.5% MoM, above typical (+6.5% MoM) & 135% YoY on continued memory strength


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On Thursday the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including June 2026.

Total semiconductor sales grew 135% YoY in June after rising 128% in May. YoY strength was again dominated by memory with sales more than quadrupling at 379% YoY, driving ~ $275B in incremental revenue YTD and with memory pricing alone driving ~2/3 of overall industry growth since the beginning of the year. Excluding memory, sales were still up strongly at 38% YoY.

Total semiconductor sales MoM were above typical seasonality, up 10.5% versus the historical June average of +6.5% MoM. On a MoM basis as-reported, product group performance was generally better than seasonal patterns.

Discretes (18.4% vs. typical 17.1%), Sensors & Actuators (12.7% vs. typical 8.3%), Analog Std Linear (14.6% vs. typical 8.9%), Analog App Specific (15.9% vs. typical 10.6%), Logic (7.5% vs. typical 4.3%), MPU (23.2% vs. typical 5.2%), DRAM (7.6% vs. typical 5.9%), NAND (14.8% vs. typical 10.2%), were better than typical MoM patterns. MCU (12.6% vs. typical 18.7%), DSP (11.6% vs. typical 15.9%), were below typical MoM patterns, while Optoelectronics (9.1% vs. typical 8.4%), were in line with historical MoM patterns.

Total unit shipments were up 11.7% MoM, with ASPs down -1.1% MoM.

By geography, total sales increased YoY and MoM in all regions.

Application specific ICs sales were up (10.6%) MoM, better than typical historical patterns (up 6.6% MoM). 2 end market were better than typical MoM, 1 was worse than typical, and 3 were inline with typical. Consumer (10.8% vs. typical 6.0%), Computer & Peripherals (13.1% vs. typical 4.3%) were above typical. Wired Communication (2.7% vs. typical 2.3%), Automotive (14.6% vs. typical 14.1%), Multipurpose and Others (12.9% vs. typical 13.8%) were in-line with typical. Wireless Communication (5.3% vs. typical 6.7%) was worse than typical.

BERNSTEIN TICKER TABLE

			6 Aug 2026		TTM		Reported EPS			Reported P/E (x)		
Ticker	Rating	Cur	Closing Price	Price Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
AMD (Advanced Micro)	O	USD	489.28	650.00	178.4%	USD	4.17	7.55	15.56	117.2	64.8	31.4
ADI (Analog Devices)	M	USD	377.31	430.00	49.5%	USD	7.79	12.40	14.65	48.4	30.4	25.8
AVGO (Broadcom)	O	USD	420.57	550.00	17.9%	USD	6.82	11.60	18.69	61.7	36.3	22.5
INTC (Intel)	M	USD	99.81	110.00	367.5%	USD	0.43	1.55	1.81	234.1	64.5	55.0
NVDA (NVIDIA)	O	USD	218.99	315.00	0.5%	USD	4.77	9.19	12.52	45.9	23.8	17.5
NXPI (NXP Semiconductors)	M	USD	231.78	290.00	(9.0)%	USD	11.81	15.02	16.85	19.6	15.4	13.8
QCOM (Qualcomm)	M	USD	160.39	165.00	(11.5)%	USD	12.03	10.51	9.57	13.3	15.3	16.8
TXN (Texas Instruments)	M	USD	278.40	290.00	28.2%	USD	5.45	8.33	9.16	51.1	33.4	30.4
KLAC	O	USD	193.22	225.00	96.0%	USD	3.76	5.47	6.52	51.4	35.3	29.6
LRCX (Lam Research)	O	USD	305.77	360.00	197.2%	USD	5.82	9.17	10.14	52.6	33.4	30.1
AMAT (Applied Materials)	O	USD	527.48	525.00	174.6%	USD	9.42	12.17	15.56	56.0	43.4	33.9
TSM (TSMC)	O	USD	418.20	430.00	59.2%	USD	10.48	16.07	19.70	12.7	9.4	7.2
2330.TT (TSMC)	O	TWD	2,365.00	2,780.00	70.9%	TWD	66.25	101.62	124.63	11.3	8.4	6.4
2303.TT (UMC)	U	TWD	121.50	88.00	171.2%	TWD	3.34	4.58	5.56	4.0	3.6	3.4
5347.TT (Vanguard)	M	TWD	151.50	146.00	28.7%	TWD	4.30	5.70	6.59	4.4	4.1	3.9
2454.TT (MediaTek)	O	TWD	3,920.00	4,380.00	159.7%	TWD	66.17	69.52	170.50	59.2	56.4	23.0
3034.TT (Novatek)	M	TWD	543.00	480.00	(7.5)%	TWD	26.87	30.43	32.54	20.2	17.8	16.7
005930.KS (SEC- Samsung)	O	KRW	230,000	440,000	196.7%	KRW	6,611.53	48,393	77,273	34.8	4.8	3.0
005935.KS (SEC-Pref - Samsung)	O	KRW	172,100	374,000	169.2%	KRW	6,611.53	48,393	77,273	26.0	3.6	2.2
SMSN.LI (Samsung)	O	USD	4,236.00	7,350.00	213.7%	USD	116.15	812.39	1,290.80	36.5	5.2	3.3
000660.KS (SK hynix)	O	KRW	1,496,000	3,300,000	441.4%	KRW	60,341	395,677	568,862	24.8	3.8	2.6
MU (Micron)	O	USD	881.47	1,300.00	688.8%	USD	8.29	67.39	158.99	106.4	13.1	5.5
285A.JP (KIOXIA)	U	JPY	48,740	40,000	1989.6%	JPY	1,014.00	10,013	9,656.84	48.1	4.9	5.0
688012.CH (AMEC)	O	CNY	361.59	500.00	145.0%	CNY	3.40	4.95	7.18	106.3	73.1	50.3
688041.CH (Hygon)	O	CNY	291.05	450.00	83.5%	CNY	1.10	2.04	3.59	264.6	142.4	81.0
002371.CH (NAURA)	O	CNY	742.50	680.00	93.5%	CNY	5.66	10.22	16.41	131.1	72.7	45.3
688072.CH (Piotech)	O	CNY	682.03	580.00	281.2%	CNY	3.32	8.12	12.40	205.4	84.0	55.0
6415.TT (Silergy)	O	TWD	445.00	600.00	3.3%	TWD	6.40	9.81	15.50	69.5	45.4	28.7
ASML.NA (ASML)	O	EUR	1,494.00	2,500.00	130.9%	EUR	24.72	38.91	53.56	60.4	38.4	27.9
ASML (ASML)	O	USD	1,704.37	2,859.00	125.2%	USD	27.95	44.01	60.58	52.9	33.6	24.4
6857.JP (Advantest)	O	JPY	32,930	45,800	187.8%	JPY	513.30	948.96	1,088.87	64.2	34.7	30.2
6146.JP (DISCO)	O	JPY	61,590	99,000	15.6%	JPY	1,245.79	1,830.28	2,336.94	49.4	33.7	26.4
7741.JP (Hoya)	O	JPY	25,475	34,800	(1.8)%	JPY	744.42	832.21	966.52	34.2	30.6	26.4
4062.JP (Ibiden)	O	JPY	19,010	29,000	432.2%	JPY	214.91	329.25	511.84	88.5	57.7	37.1
6525.JP (Kokusai)	O	JPY	8,224.00	12,420	122.6%	JPY	128.63	262.29	373.69	63.9	31.4	22.0
6920.JP (Lasertec)	O	JPY	43,330	54,000	165.2%	JPY	863.65	1,185.31	1,348.53	50.2	36.6	32.1
6723.JP (Renesas)	O	JPY	3,673.00	6,300.00	75.6%	JPY	181.61	261.36	302.64	20.2	14.1	12.1
3436.JP (SUMCO)	M	JPY	3,544.00	3,290.00	154.9%	JPY	(33.61)	(40.64)	70.73	(105.4)	(87.2)	50.1
8035.JP (Tokyo Electron)	O	JPY	55,330	79,300	125.3%	JPY	1,250.88	1,687.17	2,635.81	44.2	32.8	21.0
SOI.FP (Soitec)	O	EUR	120.75	200.00	210.4%	EUR	(6.17)	(0.32)	3.59	(19.6)	(379.6)	33.6
XFAB.FP (X-Fab)	O	EUR	6.77	12.80	(24.1)%	USD	0.23	0.07	0.51	20.1	52.4	15.9
7735.JP (Screen)	M	JPY	13,690	15,000	95.7%	JPY	486.61	574.49	759.55	28.1	23.8	18.0
SPX			7,709.96									
ASIAX			1,909.27									
EM			1,778.30									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, KLAC, LRCX, 6920.JP, SOI.FP base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

Ticker	Rating	Cur	6 Aug 2026 Closing Price	Price Target	TTM Rel. Perf.	Reported EPS Cur	2025A	2026E	2027E	Reported P/E (x) 2025A	2026E	2027E
EDME			1,633.76									
JPL			2,628.52									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP estimate is Adjusted EPS; AMD, ADI, AVGO, INTC, NVDA, NXPI, QCOM, KLAC, LRCX, AMAT, MU, 6723.JP, SOI.FP valuation is Adjusted P/E (x); TSM, 2330.TT, 2303.TT, 5347.TT valuation is P/B (x); XFAB.FP valuation is EV/EBIT (x); NVDA, KLAC, LRCX, 6920.JP, SOI.FP base year is 2026;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

ADI (Market-Perform, $430.00): ADI is executing well with both cyclical and idiosyncratic drivers, though may be closer to peak, and valuations remain elevated.

AMD (Outperform, $650.00): Expectations remain high, but exposure to AI demand driving both a CPU and GPU story can provide substantial growth.

AVGO (Outperform, 550.00): A strong 2026 AI trajectory seems set to accelerate into 2027 and beyond, bolstered by software, cash deployment, and superb margins & FCF.

INTC (Market-Perform, $110.00): Server strength is helping the company get back on their feet, and narrative/headlines may fuel the vibe for now.

NVDA (Outperform, $315.00): The datacenter opportunity is enormous, and still early, with material upside still possible.

NXPI (Market-Perform, $290.00): The pace and makeup of recovery remains open for debate, though company execution is solid and valuation is increasingly inexpensive.

QCOM (Market-Perform, $165.00): The diversification story is real but a number of near-term headwinds (smartphone memory dynamics, margins, AAPL etc) need to be navigated first.

TXN (Market-Perform, $290.00): TXN shares feel fully valued in the current environment.

AMAT (Outperform, $525.00): We maintain a positive view on secular WFE growth and see a number of drivers for AMAT including SAM growth, an increasing services narrative, and capital return.

LRCX (Outperform, $360.00): The company is benefiting from key inflections (GAA, packaging, HBM, NAND upgrades) and CY26/CY27 commentary seems supportive.

KLAC (Outperform, **$225.00**): Amid positive WFE trends KLAC possess structural growth drivers, a strong and durable competitive position, lower China replacement risk, and disciplined capital allocation, warranting premium valuation.

TSMC: We rate TSMC Outperform with PT=NT$2,780.00.

UMC: We rate UMC Underperform with PT=NT$88.00.

Vanguard: We rate Vanguard Market-Perform with PT=NT$146.00.

Novatek: We rate Novatek Market-Perform with PT=NT$480.00.

MediaTek: We rate MediaTek Outperform with PT=NT$4,380.00.

Samsung Electronics: We rate Samsung Electronics Outperform with target price of KRW 440,000.

SK hynix: We rate SK hynix Outperform with target price to KRW 3,300,000.

Micron: We rate Micron Outperform with target price of US$1,300.00.

KIOXIA: We rate KIOXIA Underperform with target price of JPY40,000.

SOI.FP: We rate Soitec Outperform with target price of €200.00.

XFAB.FP: We rate X-FAB Outperform with target price of €12.80

NAURA: We rate Naura Outperform with PT of CNY 680.00.

AMEC: We rate AMEC Outperform with PT of CNY 500.00.

Piotech: We rate Piotech Outperform with PT of CNY 580.00.

Hygon: We rate Hygon Outperform with PT of CNY 450.00.

Silergy: We rate Silergy Outperform with PT of TWD 600.00.

DISCO: We rate DISCO Outperform with PT of JPY 99,000.

Tokyo Electron: We rate Tokyo Electron Outperform with PT of JPY 79,300.

ASML: We rate ASML Outperform with PT of EUR 2,500.00.

Advantest: We rate Advantest Outperform with PT of JPY 45,800.

Screen: We rate Screen Market-Perform with PT of JPY 15,000.

Kokusai: We rate Kokusai Outperform with PT of JPY 12,420.

Lasertec: We rate Lasertec Outperform with PT of JPY 54,000.

SUMCO: We rate SUMCO Market-Perform with PT of JPY 3,290.00.

Hoya: We rate Hoya Outperform with PT of JPY 34,800.

Ibiden: We rate Ibiden Outperform with PT of JPY 29,000.

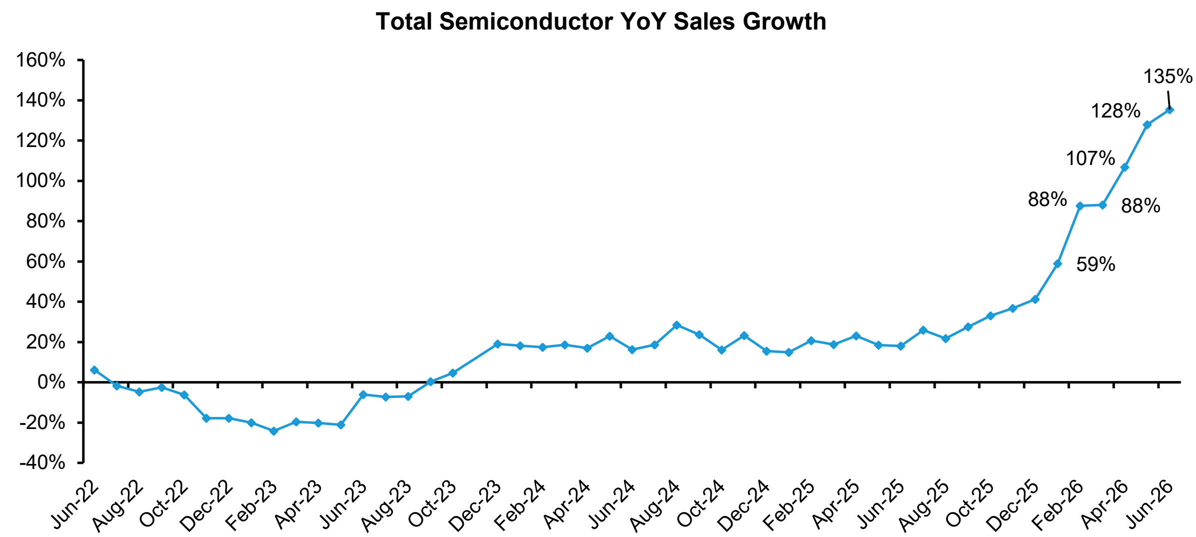
Renesas: We rate Renesas Outperform with PT of JPY 6,300.00.

DETAILS

On Thursday the SIA published its monthly WSTS report, which tracks sales, units and ASPs across the semiconductor industry, and provides details by products, end markets, and geographies. The recent report includes data up to and including June 2026.

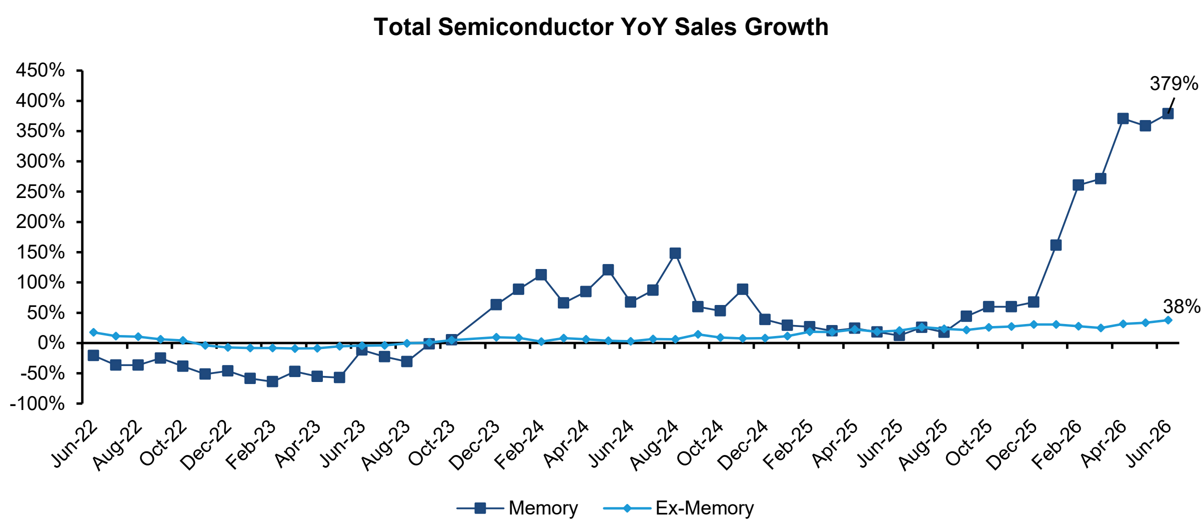
Total semiconductor sales grew 135% YoY in June after rising 128% in May (**Exhibit 1**). YoY strength was again dominated by memory with sales more than quadrupling at 379% YoY, driving ~$275B in incremental revenue YTD and with memory pricing alone driving ~2/3 of overall industry growth since the beginning of the year (**Exhibit 3, Exhibit 4**). Excluding memory, sales were still up strongly at 38% YoY (**Exhibit 2**). On a 3-month rolling basis sales were up 35.1%, much better than typical historical patterns of up MSD (4.6%) (**Exhibit 5**).

EXHIBIT 1: **Historical total semiconductor YoY sales growth**



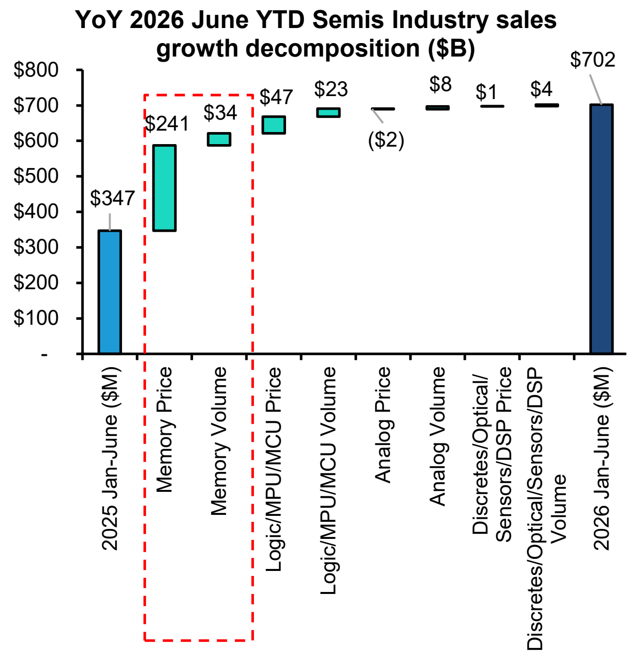
Source: WSTS, Bernstein Analysis

EXHIBIT 2: **Historical memory vs. non-memory YoY sales growth**



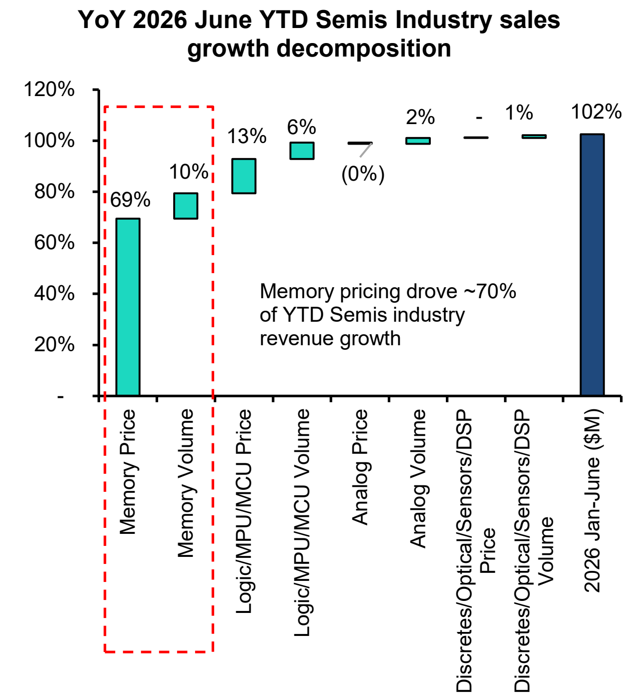
Source: WSTS, Bernstein Analysis

EXHIBIT 3: **Memory drove ~$275B in incremental Semis sales YTD through June...**



Price x volume mix effects attributed to price.
Source: WSTS, Bernsetin analysis

EXHIBIT 4: **...with Memory pricing alone driving ~70% of YTD Semis revenue growth**



Price x volume mix effects attributed to price.
Source: WSTS, Bernstein analysis

EXHIBIT 5: **Semiconductor industry % change in sales**

% Change in Sales

	vs 1 month ago		vs 3 months ago		3 months rolling		vs 1 year ago
	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This Month	Historical Average (20-25)	This month
Discretes	18.4%	17.1%	2.5%	0.0%	7.2%	2.0%	10.8%
Optoelectronics	9.1%	8.4%	1.3%	-6.2%	3.8%	-4.1%	16.4%
Sensors & Actuators	12.7%	8.3%	1.2%	-1.3%	3.2%	0.7%	6.9%
Analog Std Linear	14.6%	8.9%	6.5%	-1.6%	9.0%	3.4%	41.5%
Analog App Specific	15.9%	10.6%	1.6%	-2.0%	-0.2%	-1.3%	12.5%
Logic	7.5%	4.3%	11.7%	3.8%	12.2%	4.3%	50.4%
MPU	23.2%	5.2%	29.1%	0.5%	16.1%	1.9%	49.3%
MCU	12.6%	18.7%	6.1%	3.0%	8.6%	1.0%	18.1%
DSP	11.6%	15.9%	19.0%	-3.5%	25.6%	-1.6%	-4.7%
Memory	10.1%	7.0%	50.5%	9.3%	64.2%	15.0%	378.9%
DRAM	7.6%	5.9%	45.5%	6.3%	58.4%	14.2%	373.3%
NAND	14.8%	10.2%	60.2%	16.8%	77.0%	18.0%	406.6%
TOTAL	10.5%	6.5%	30.7%	2.4%	35.1%	4.6%	135.4%

Source: WSTS, Bernstein Analysis

Total semiconductor sales MoM were above typical seasonality, up 10.5% versus the historical June average of 6.5% MoM. On a MoM basis as-reported, product group performance was generally better than seasonal patterns.

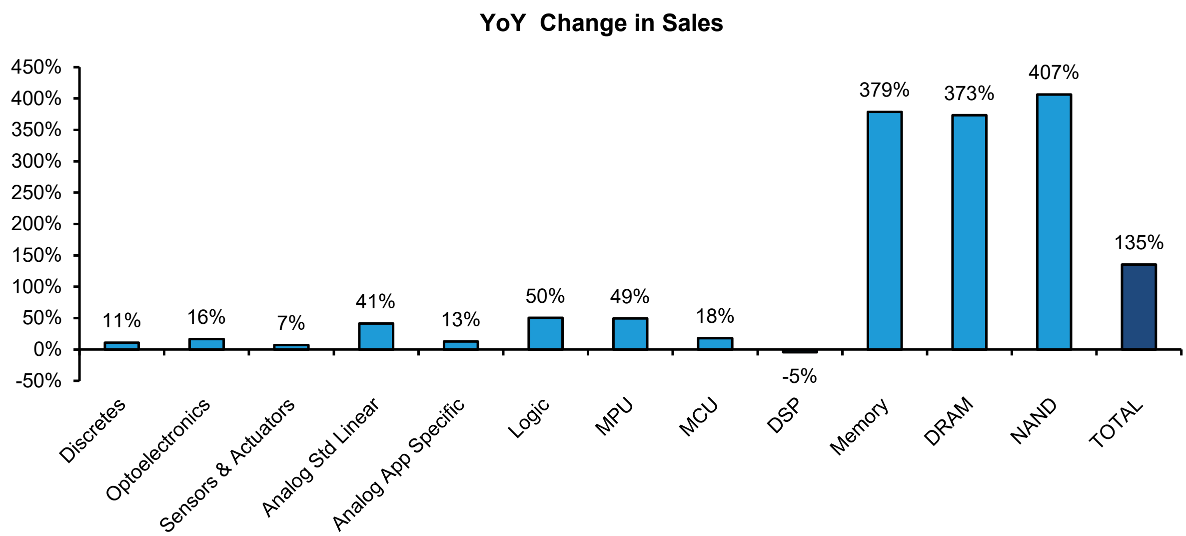
In June on a YoY basis and on a product group level, all product groups were up, except DSPs (which were down 4.7% YoY) (**Exhibit 6**).

On a MoM basis, performance by product group was relatively positive with 8 groups better than typical, 1 in line, and 2 groups worse than typical patterns. (**Exhibit 7**).

- Discretes (18.4% vs. typical 17.1%), Sensors & Actuators (12.7% vs. typical 8.3%), Analog Std Linear (14.6% vs. typical 8.9%), Analog App Specific (15.9% vs. typical 10.6%), Logic (7.5% vs. typical 4.3%), MPU (23.2% vs. typical 5.2%), DRAM (7.6% vs. typical 5.9%), NAND (14.8% vs. typical 10.2%), were better than typical MoM patterns.

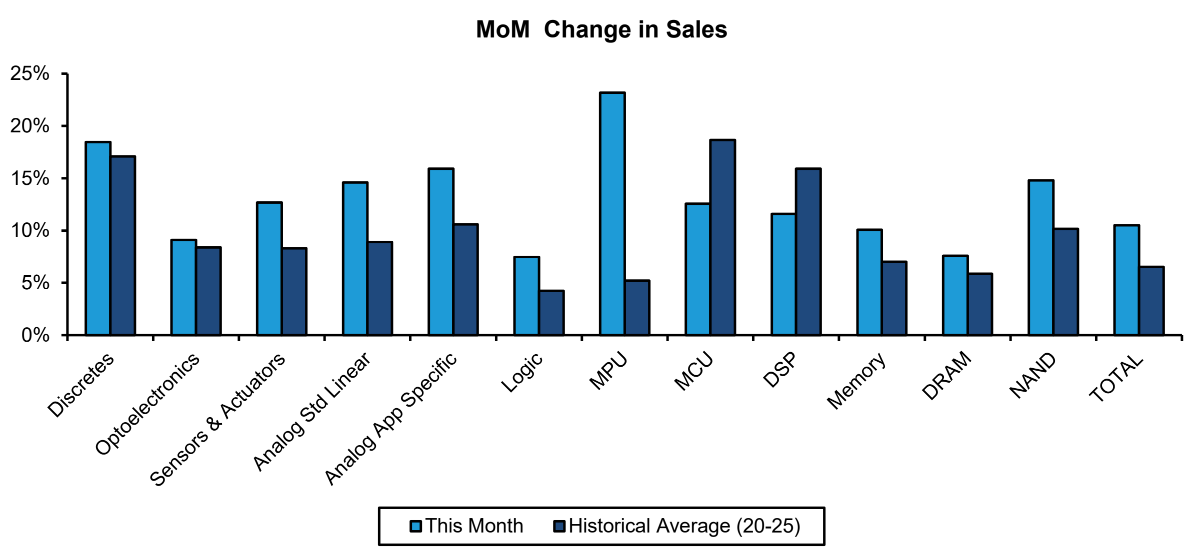
- MCU (12.6% vs. typical 18.7%), DSP (11.6% vs. typical 15.9%), were below typical MoM patterns, while Optoelectronics (9.1% vs. typical 8.4%), were in line with historical MoM patterns.

EXHIBIT 6: **Total semiconductor YoY change in sales**



Source: WSTS, Bernstein Analysis

EXHIBIT 7: **Total semiconductor MoM change in sales**



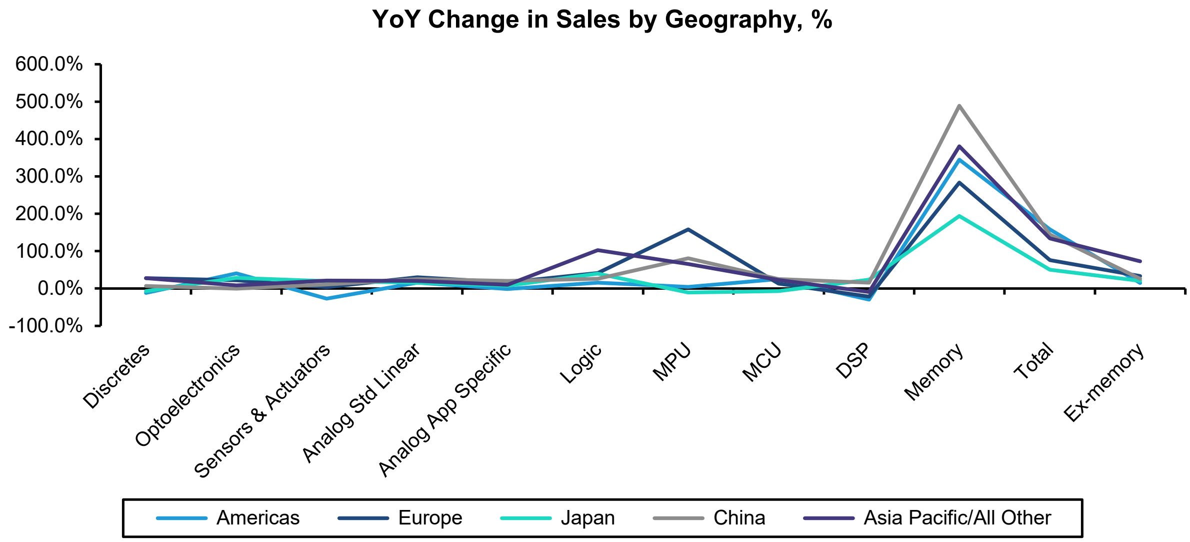
Source: WSTS, Bernstein Analysis

By geography, total sales increased YoY and MoM in all regions.

On a YoY basis, total semiconductors sales were up 157.9% in the Americas, 76.3% in Europe, 49.9% in Japan, 144.6 in China, and 133.7% in Asia Pacific/All Other **(Exhibit 8)**.

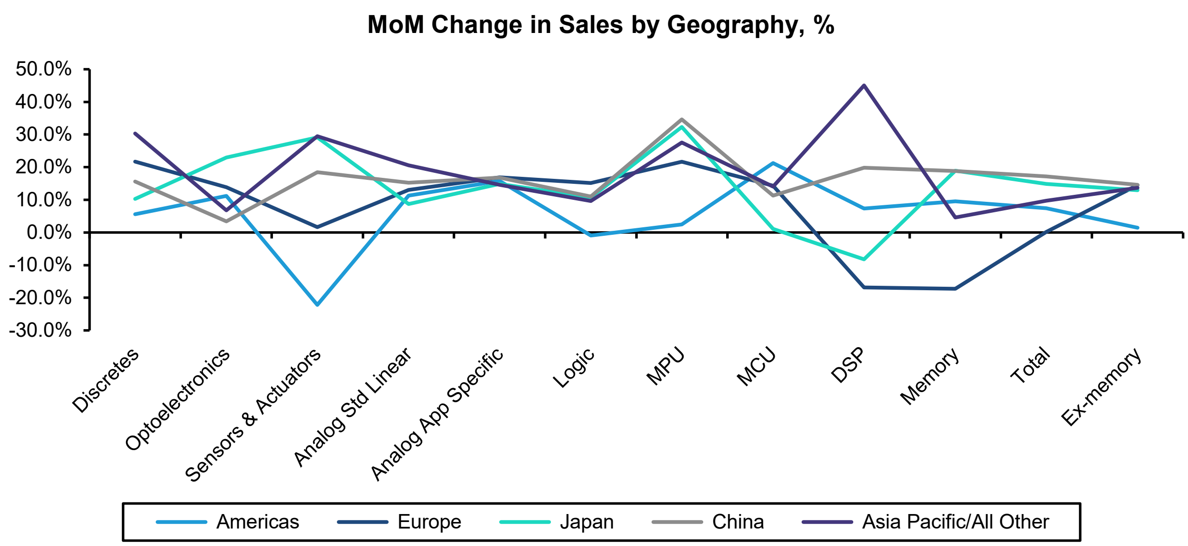
On a MoM basis, sales were up 7.4% in the Americas, 0.2% in Europe, 14.9% in Japan, 17.2% in China, and 9.8% in Asia Pacific/All Other **(Exhibit 9)**.

EXHIBIT 8: **Total semiconductor YoY change in sales by geography**



Source: WSTS, Bernstein Analysis

EXHIBIT 9: **Total semiconductor MoM change in sales by geography**



Source: WSTS, Bernstein Analysis

Total unit shipments were up 11.7% MoM, with ASPs down -1.1% MoM.

- Unit shipments were up MoM for 11 product groups.
- Units were up MoM for Discretes, Optoelectronics, Sensors & Actuators, Analog Std Linear, Analog App Specific, Logic, MPU, MCU, DSP, DRAM[bits], NAND[bits], (10.5%, 7.8%, 18.6%, 16.9%, 20.1%, 8.7%, 25.5%, 9.2%, 14.7%, 3.5%, 10.7% respectively).
- On a 3 months rolling basis overall shipments were up 9.5%.

- On a YoY basis shipments increased 8.3%. Shipments declined for Discretes, and units increased YoY for all other product groups (Optoelectronics, Sensors & Actuators, Analog Std Linear, Analog App Specific, Logic, MPU, MCU, DSP, DRAM[bits], NAND[bits]). (**Exhibit 10**).

EXHIBIT 10: **Change in semiconductor unit shipments**

% Change in Units

	vs 1 month ago	vs 3 months ago	3 months rolling	vs 1 year ago
Discretes	10.5%	3.7%	8.6%	-4.8%
Optoelectronics	7.8%	13.1%	5.9%	16.8%
Sensors & Actuators	18.6%	4.7%	6.3%	9.7%
Analog Std Linear	16.9%	10.4%	12.6%	23.3%
Analog App Specific	20.1%	-4.1%	3.1%	15.3%
Logic	8.7%	7.9%	12.9%	20.1%
MPU	25.5%	22.5%	3.5%	21.5%
MCU	9.2%	3.2%	3.7%	14.9%
DSP	14.7%	30.1%	31.8%	16.9%
DRAM (Based on Bit Shipped)	3.5%	6.9%	11.7%	26.1%
NAND (Based on Bit Shipped)	10.7%	7.3%	7.2%	8.2%
TOTAL	11.7%	6.6%	9.5%	8.3%

Source: WSTS, Bernstein Analysis

On a MoM basis, industry ASPs were down -1.1%, with 5 out of 11 product groups experiencing ASP increases and 6 seeing ASP decreases

- ASPs for Discretes, Optoelectronics, MCU, DRAM[bits], NAND[bits], saw increases (6.0%, 1.2%, 3.1%, 4.0%, 3.7% respectively).

- ASPs fell MoM for Sensors & Actuators, Analog Std Linear, Analog App Specific, Logic, MPU, DSP, (-5.0%, -2.1%, -3.4%, -1.1%, -1.9%, -2.7%, respectively).

- Versus 3 months ago, 6 out of 11 product groups saw ASP increases, and 5 seeing decreases, while on a YoY basis 7 out of 11 groups saw increases, 1 flattish, and 3 down . Total industry ASPs rose 117.5 YoY on the back of memory price increases. (**Exhibit 11**).

EXHIBIT 11: **Change in semiconductor ASPs**

% Change in ASP

	vs 1 month ago	vs 3 months ago	vs 1 year ago
Discretes	6.0%	-1.4%	16.4%
Optoelectronics	1.2%	-10.3%	-0.4%
Sensors & Actuators	-5.0%	-3.4%	-2.6%
Analog Std Linear	-2.1%	-3.7%	14.5%
Analog App Specific	-3.4%	6.0%	-2.3%
Logic	-1.1%	3.5%	25.2%
MPU	-1.9%	5.4%	22.9%
MCU	3.1%	2.9%	2.8%
DSP	-2.7%	-8.5%	-18.5%
DRAM (Based on $/Bit)	4.0%	36.1%	275.3%
NAND (Based on $/Bit)	3.7%	49.3%	368.4%
TOTAL	-1.1%	22.7%	117.5%

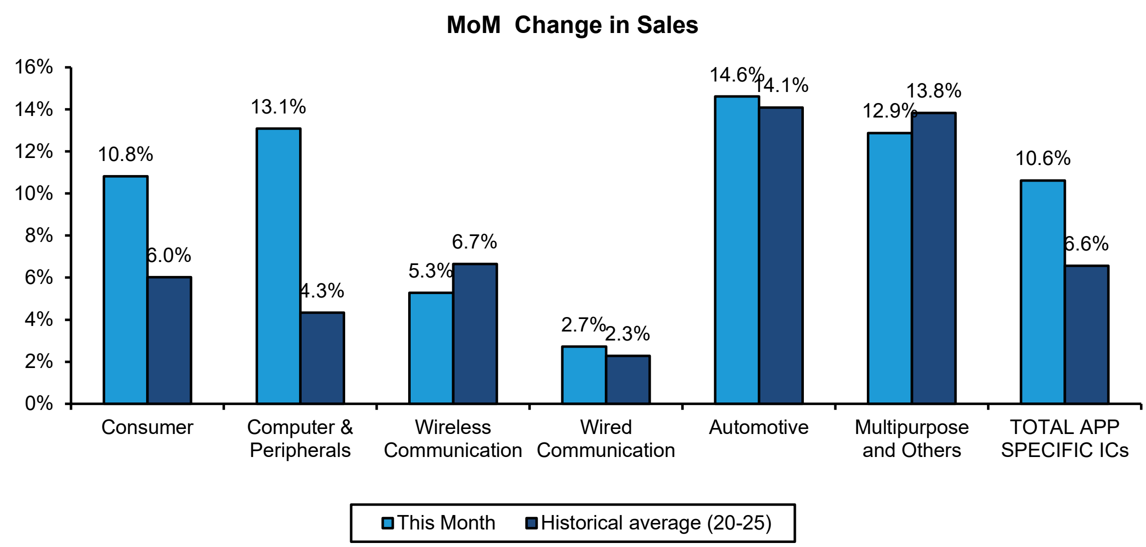
Source: WSTS, Bernstein Analysis

Application specific ICs sales were up (10.6%) MoM, better than typical historical patterns (up 6.6%) MoM. 2 end market were better than typical MoM, 1 was worse than typical, and 3 were inline with typical (Exhibit 12).

- Application specific ICs sales were up (10.6%) MoM, better than typical historical patterns (6.6% MoM).

- 2 product groups were above typical MoM: Consumer (10.8% vs. typical 6.0%), Computer & Peripherals (13.1% vs. typical 4.3%).
- 1product group was worse than typical: Wireless Communication (5.3% vs. typical 6.7%).
- 3 product groups were in-line with typical: Wired Communication (2.7% vs. typical 2.3%), Automotive (14.6% vs. typical 14.1%), Multipurpose and Others (12.9% vs. typical 13.8%).

EXHIBIT 12: **Change in sales by end markets (where WSTS data provides a split by application)**



Source: WSTS, Bernstein Analysis

By product category, we note the following:

Analog Standard Linear: Total Analog Standard Linear sales were up (14.6%), better than typical patterns (8.9%). Shipments were up (16.9%) with ASPs down (-2.1%). Product category changes MoM were as follows: Amplifiers/Comparators (12.1% vs. typical 10.4%), Interface (19.6% vs. typical 13.0%), Power Management (19.1% vs. typical 8.3%), and Signal Conversion (-3.9% vs. typical 8.0%) (**Exhibit 13**).

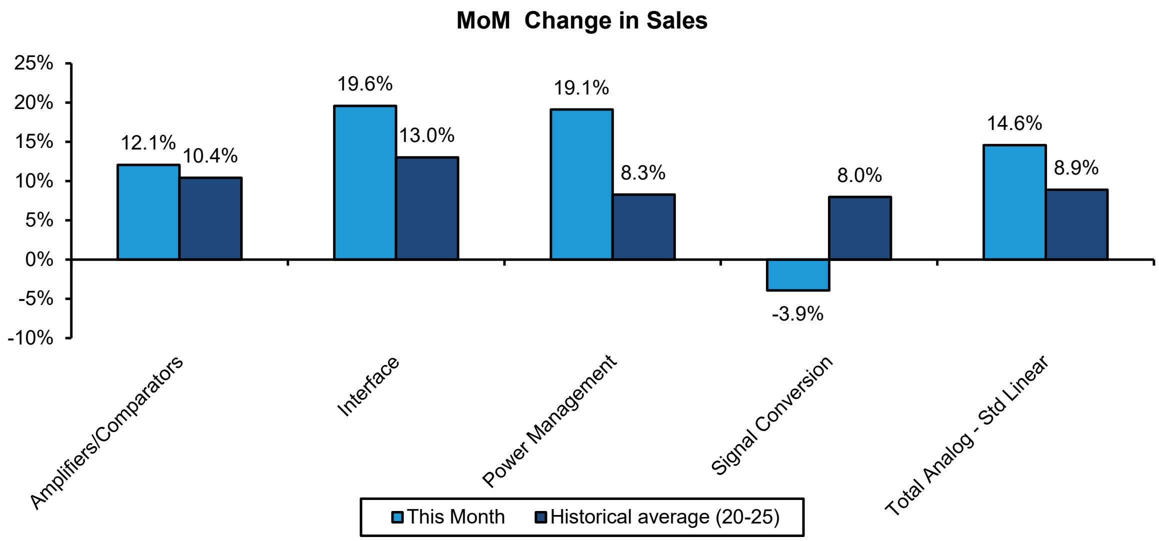
Analog Application Specific: Total analog application specific sales were up (15.9%) MoM, better than typical of 10.6% MoM. Units were up (20.1%) MoM, with ASPs down (-3.4%). End markets changes MoM were as follows: Consumer (28.1% vs. typical 16.1%), Computer (25.9% vs. typical 8.3%), Communications (11.7% vs. typical 6.5%), Automotive (14.3% vs. typical 15.1%), and Industrial & Others (22.5% vs. typical of 20.5%) (**Exhibit 14**).

MPU: MPU sales increased 23.2% MoM, better than typical patterns (5.2%). Units were up (25.5%) MoM with ASPs down (-1.9%) MoM. On a 3-months rolling basis (e.g. April-June vs. January-March), MPU sales increased (16.1%), better than typical average of 1.9%. Sales increased 29.1% vs. 3 months ago and increased 49.3% YoY.

MCU: MCU sales were up (12.6%) MoM, worse than historical averages of up 18.7%, with units up (9.2%) and ASP up (3.1%). Consumer was up 7.0% MoM vs. typical 27.7% MoM, Computer was 2.6% MoM vs. typical 17.8% MoM, Handsets were down -16.5% MoM vs. typical of up 53.9% MoM, Wireless Communication-All Other was up 34.4% MoM, Wired Communication was up 49.9% MoM vs. typical of 27.1% MoM, Automotive was up 10.1% MoM vs. typical 14.9% MoM, and Multi-purpose and others were 22.4% MoM vs. typical of 22.0% MoM. (**Exhibit 15**).

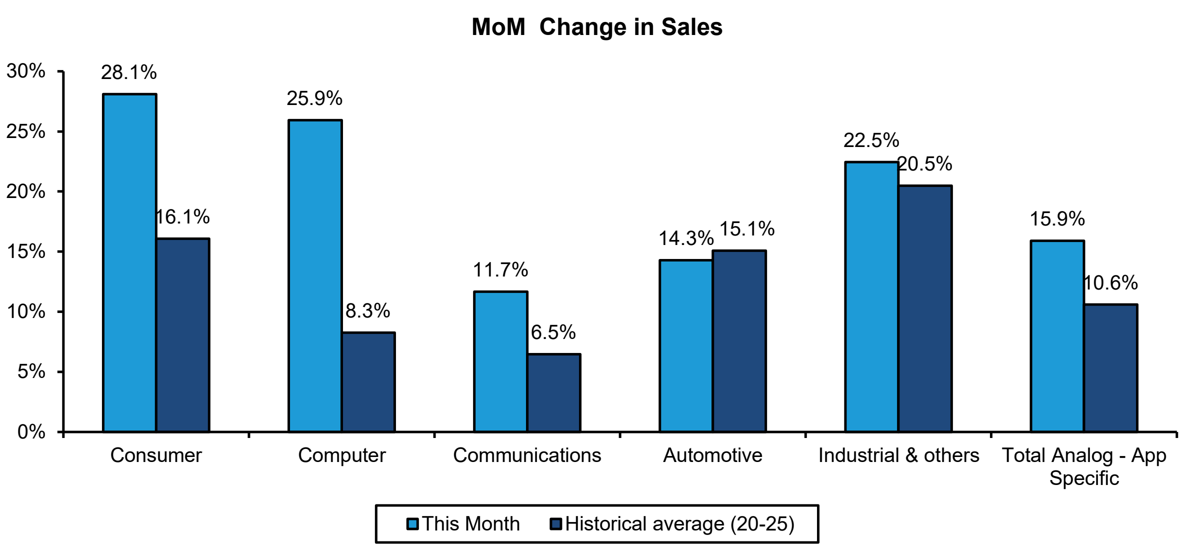
Memory: Memory sales were up (10.1%) MoM, better than historical average of up 7.0%. On a bit shipped basis, DRAM shipments were up 3.5% MoM and NAND up 10.7% MoM. On a $ per bit basis, DRAM saw ASPs up 4.0%, while NAND ASPs up 3.7% (**Exhibit 16**, **Exhibit 17**).

EXHIBIT 13: **Analog Standard Linear - MoM change in sales**



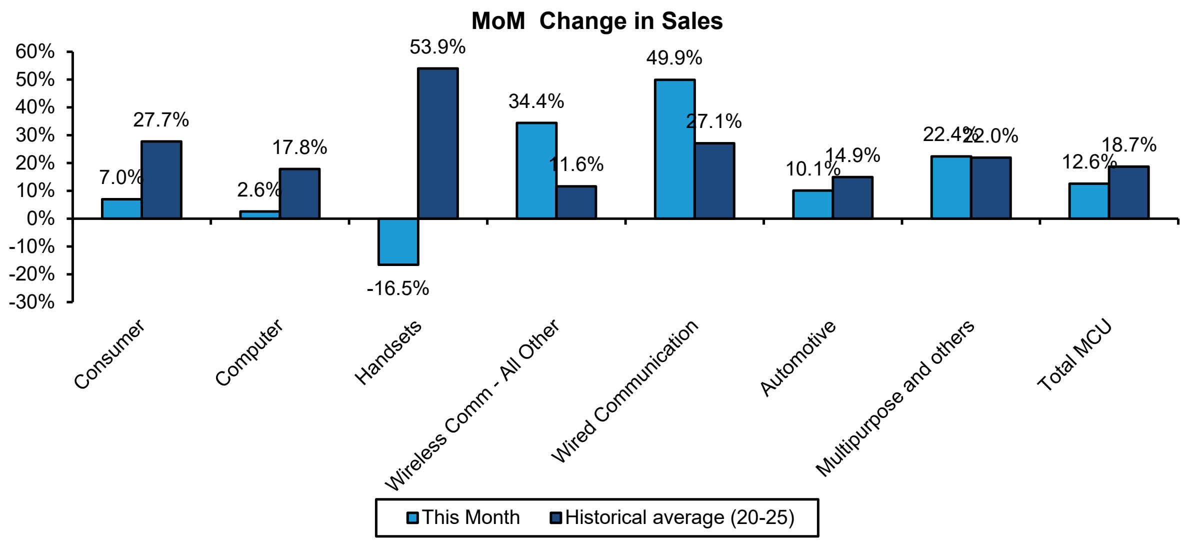
Source: WSTS, Bernstein Analysis

EXHIBIT 14: **Analog App Specific - MoM change in sales**



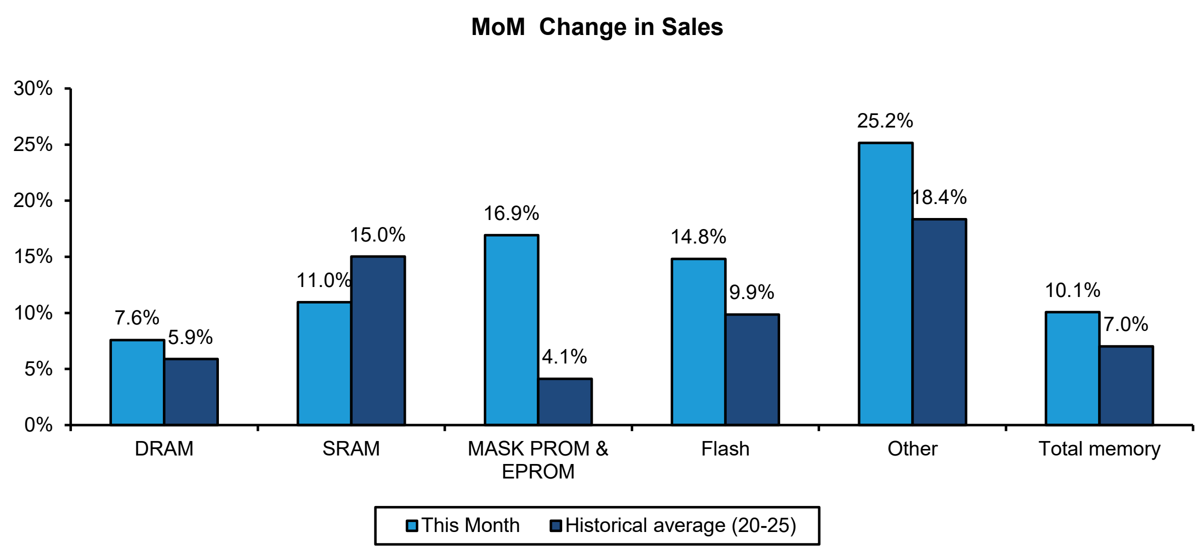
Source: WSTS, Bernstein Analysis

EXHIBIT 15: **MCU - MoM change in sales**



Source: WSTS, Bernstein Analysis

EXHIBIT 16: **Memory - MoM change in sales**



Source: WSTS, Bernstein Analysis

EXHIBIT 17: **Memory - Change in units and ASP on bit basis**

% change based on bit shipped

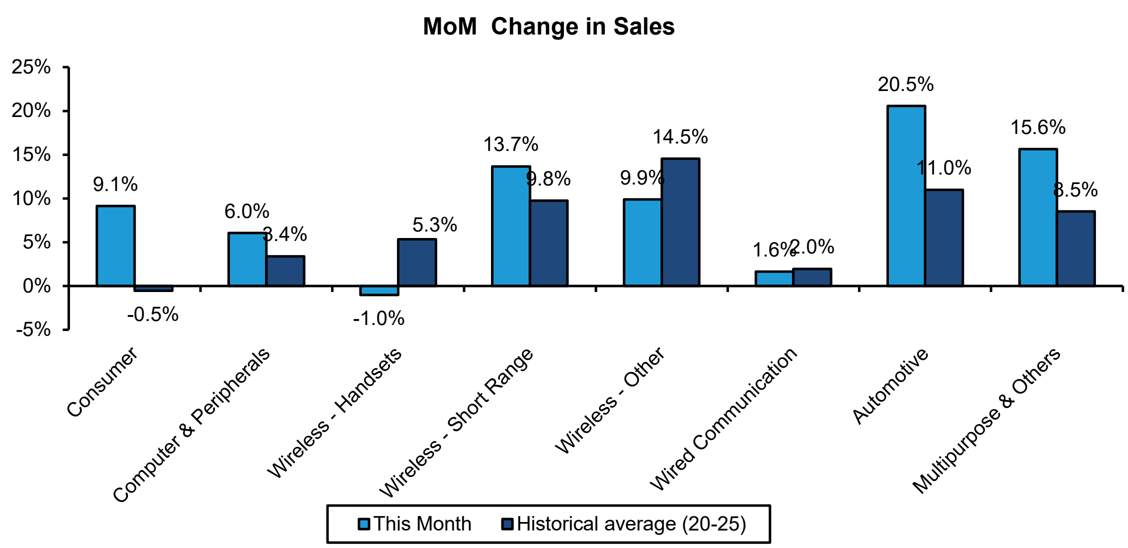
	vs 1 month ago	vs 3 months ago	3 months rolling	vs 1 year ago
DRAM	3.5%	6.9%	11.7%	26.1%
NAND	10.7%	7.3%	7.2%	8.2%

% change in $ per bit

	vs 1 month ago	vs 3 months ago	vs 1 year ago
DRAM	4.0%	36.1%	275.3%
NAND	3.7%	49.3%	368.4%

Source: WSTS, Bernstein Analysis

EXHIBIT 18: **MOS Special Purpose Logic - MoM change in sales**



Source: WSTS, Bernstein Analysis

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